Answer RFIs before they become delays

RFI Copilot for Summit Ridge Builders · Engineering & Construction · General contractor, 8 active and recent projects, $995M contract value

About $1.6M a year in protected margin. RFI response time cut from 7.9 to 6 days. RFIs likely to miss their SLA are flagged the day they are submitted, before the delay reaches the critical path.

Built end to end on the Databricks Data Intelligence Platform. All figures come from the working prototype running on synthetic but realistic portfolio data.

The outcome, in your KPIs

KPI (portfolio)	Today	Target with RFI Copilot	Who feels it
Average days to answer an RFI	**7.9 days** (p90: 15 days)	**≤ 6.0 days**	Superintendents, crews
RFIs that miss the contractual SLA	**46.6%**	**≤ 30%**	Project Executives
Liquidated-damages exposure from late critical-path RFIs	**$11.2M** since 2023 (≈ $3.2M/yr)	**–30%** (≈ $1.0M/yr avoided)	COO, CFO
Change-order cost originating in RFIs	**$16.8M** (740 COs)	Root-cause feedback to preconstruction	Preconstruction, Estimating
Project-engineer time per RFI	≈ $1,080 of handling cost per RFI (industry)	–35% with triage + cited AI drafts	Project Engineers

Value: about $1.6M per year for this portfolio. The levers and assumptions are on slide 8. Platform run cost is sized during the pilot (see the appendix).

The problem: RFIs are a hidden schedule risk

- **3,795 RFIs** across 8 projects. **65 are open right now, and 44 of them are already past SLA.**
- Half of all RFIs miss the design team's contractual response window (7 days, or 3 days if urgent).
- When a late RFI sits on a **critical-path** activity, every day it slips becomes a liquidated-damages day: $25K/day at Lakeside Hospital and $60K/day at Harbor Point.
- Project engineers triage by gut feel and re-type answers that already exist in the spec book. Leadership has no portfolio view of which firm, discipline or document gap is driving the cost.

> Industry context: the Navigant Construction Forum study of 1,362 projects found an average of 796 RFIs per project and 9.7 days per response.

Where it hurts most (from the gold tables and Genie)

Project	RFIs	Open / past SLA	Avg days	SLA breach	CO cost	LD exposure
Harbor Point Data Center Ph 2	778	24 / 15	10.5	66%	$3.2M	**$7.6M**
Lakeside Regional Hospital	846	8 / 4	8.5	53%	$4.2M	**$2.6M**
Union Station Mixed-Use Tower	537	6 / 5	7.3	42%	$2.2M	$0.8M

- **Design firms:** the MEP and fire-protection engineers breach SLA 57–65% of the time, versus 21–36% for civil and structural.
- **Root cause (Gen AI on every RFI):** spec ambiguity is 28% of volume, and field conditions plus design conflicts drive **$10.3M** of change-order cost.
- **Implication:** two projects and two design firms account for most of the exposure. That is where to start.

What changes for the project team (domain owner)

Today: an RFI arrives by email or in Procore. The PE reads it, searches the spec PDF, types an answer, forwards it to the architect and chases it in a spreadsheet.

With RFI Copilot:

1. **One inbox ranked by risk.** Every open RFI shows the ML probability that the design team will miss the SLA, the expected days to answer, and *why* (for example, "MEP firm breaches more often · 26 RFIs on this project in 14 days · no drawing attached").
2. **Triage gate.** If the spec already answers it, the PE closes it in one click and it never reaches the design team. About 28% of RFIs are spec-ambiguity questions.
3. **A cited draft is waiting.** Vector Search finds the governing spec section, and Claude drafts a response that cites it (for example, "per 21 13 13"). The PE edits and approves it.
4. **Approver sign-off (human in the loop).** High-risk, urgent or cost/schedule-impacting responses stop in an Approvers queue for a named Project Executive, who signs off or returns them with a comment. Low-risk answers go straight to the design firm.
5. **Route, flag a PCO, close.** Every action shows a live trace (Lakebase commit in about 5 ms, then the audit job merging into Delta) and lands in the RFI's audit trail.
6. **Ask Genie.** "Which open RFIs on Harbor Point will miss SLA?" returns a plain-English answer plus the SQL behind it.

How it works: one integrated data journey

1. **Lakeflow:** Auto Loader ingests the nightly PM-system exports (RFI logs as JSON, design-firm SLAs as CSV, the spec book as Markdown) into bronze, silver and gold tables. Expectations removed 49 defective or duplicate records in the latest run.
2. **Unity Catalog:** tags, descriptions, a column mask that hides change-order dollars from anyone outside finance, least-privilege grants to the app, and automatic lineage (19 table-to-table edges captured).
3. **ML:** an SLA-breach classifier and a days-to-answer model, tracked in MLflow, registered in Unity Catalog as `@champion`, and run nightly to score every open RFI.
4. **Gen AI:** `ai_classify` assigns a root cause to every RFI. A Vector Search index over the spec book plus Claude (Foundation Model API) produces cited drafts.
5. **Lakebase:** a synced table serves the inbox to the app in about 6 ms. App decisions are Postgres transactions that a Lakeflow Job merges back into a governed Delta audit table.
6. **Genie + Databricks App:** a Genie space over the gold, ML, root-cause and audit tables, and the RFI Copilot app for the project team.

Everything is one Asset Bundle: `databricks bundle deploy && databricks bundle run rfi_build`.

Proof it works (prototype results)

Capability	Result from the latest end-to-end run
Lakeflow data quality	3,844 raw records → 3,795 clean: 15 unknown-project and 15 invalid-discipline rows dropped by expectations, 19 duplicate re-sends removed
ML: who will miss SLA	AUC **0.74** on a time-based holdout; **77% precision in the top-20% risk band vs a 47% base rate (1.64× lift)**
ML: days to answer	MAE 3.5 days vs 4.1 days for a naive baseline
Gen AI retrieval	The governing spec section was retrieved for **95%** of open RFIs (hit@4)
Gen AI drafts	65 of 65 drafts generated; 64 of 65 cite only spec sections present in the retrieved context
Gen AI root cause	`ai_classify` agrees with the ground-truth label for 74% of 3,795 RFIs
Lakebase serving	Point reads **p50 5.8 ms / p95 8.1 ms**; app writes commit in about 5 ms
Genie	6 of 6 executive questions answered with generated SQL (overdue by project, firm speed, design-conflict CO cost, LD exposure)

Value model (annualized for this portfolio)

Lever	Mechanism	Assumption	Annual value
Avoided liquidated damages	Escalate high-risk critical-path RFIs at submission, days before the SLA lapses	30% of the ≈ $3.2M/yr LD exposure from late critical-path RFIs avoided	**≈ $0.96M**
Project-engineer productivity	Triage gate + cited drafts	≈ 1,085 RFIs/yr × $1,080 handling cost × 35% saved	**≈ $0.41M**
Fewer RFI-driven change orders	Root-cause feedback to preconstruction (design conflicts + missing information)	10% of ≈ $1.9M/yr	**≈ $0.19M**
Crew idle time (not counted)	Faster answers on non-critical work	—	upside
Total			**≈ $1.6M / yr**

The history covers about 3.5 years (2023 to Oct 2026), so annual values are total ÷ 3.5. Every assumption is a parameter to validate during the pilot.

For the executive sponsor (COO / VP Operations)

- **Margin protection you can measure.** Liquidated damages and RFI-driven change orders are tracked per project in governed tables, so there is a weekly number, not an anecdote.
- **Early warning, portfolio-wide.** The risk model flags the RFIs most likely to stall work before the SLA lapses. Genie answers "where are we exposed?" without waiting on an analyst.
- **Design-team accountability.** Firm-level SLA performance (for example, 65% breach for one specialty engineer) gives you leverage at the next fee negotiation or selection.
- **Low-risk platform choice.** One governed platform for data, ML, Gen AI, the app database and BI. There is no new point product to integrate, and it runs on the GC's existing Databricks account.

For the domain owner (Director of Project Controls / Project Executives)

- **Less typing, same control.** AI drafts are always reviewed and approved by a person. Citations show exactly which spec section supports the answer, and gaps are flagged instead of guessed.
- **Fewer RFIs reach the design team.** The triage gate closes the ones the contract documents already answer.
- **One queue, sorted by what will hurt.** Risk tier, expected answer date, liquidated-damages rate and field proposal are all on one screen.
- **Audit trail by default.** Every approve, route, PCO flag or close is a timestamped event in Lakebase, mirrored to a governed Delta table and queryable in Genie.
- **Fits existing tools.** Ingestion reads the PM system's exports (Procore and Autodesk Construction Cloud style), so no new data entry is required.

Trust, governance and controls

- **Unity Catalog** governs every table, model, index and the synced Lakebase table: lineage from raw file to app, tags, and descriptions.
- **Least privilege:** the app runs as its own service principal with SELECT on specific tables, CAN_USE on one warehouse, CAN_RUN on the Genie space, and INSERT on one Postgres table.
- **Sensitive dollars are masked:** change-order cost is NULL for anyone outside the finance group, enforced by a column-mask function. The build's own run shows the masked output.
- **Human in the loop:** AI drafts never go out without a person approving them. A policy gate (ML risk, AI-assessed impact, urgency, LD rate) adds a named approver for high-stakes answers, and sign-off turnaround is tracked. The decision log is append-only and synced to Delta.
- **No customer data:** the prototype uses synthetic data; a pilot would run inside the GC's own workspace.

90-day pilot plan

Weeks	Scope	Success criteria
1–3	Connect Procore/ACC RFI exports and the spec book for **Harbor Point** and **Lakeside**	Daily Lakeflow ingest; gold KPIs match the PM system within 1%
4–6	Risk model retrained on the GC's own history; Copilot rolled out to 6 project engineers	Top-20% risk band precision ≥ 70%; ≥ 60% of drafts approved with light edits
7–10	Genie space for project executives; weekly risk review	Executives use Genie weekly; high-risk RFIs escalated ≥ 10 days before the SLA
11–13	Measure and decide	Average response ≤ 6 days on pilot projects; LD-exposure trend down; go/no-go for all 8 projects

The ask: a sponsor (COO), a domain owner (Director of Project Controls), two pilot projects, and read access to RFI exports and spec books.

Appendix: assumptions and sources

- Portfolio, RFIs, firms and spec book are **synthetic**, produced by `01_generate_raw_data`. The generator encodes realistic drivers: firm responsiveness, coordination load, missing drawings, priority, and a 12% critical-path rate.
- Liquidated-damages exposure = schedule-impact days × the project's LD rate, counted **only** for critical-path RFIs.
- RFI handling cost (≈ $1,080 per RFI), 796 RFIs per project and 9.7 days per response: Navigant Construction Forum, "Impact of Requests for Information on Construction Projects" (2013).
- Platform run cost (indicative, to be sized in the pilot): a nightly serverless pipeline and jobs, one small Lakebase instance, one Vector Search endpoint, a serverless SQL warehouse, and pay-per-token model calls.
- Code, executed notebooks with outputs, and app smoke-test results: see the GitHub repository (`evidence/`).